Introduction to the Smith Chart

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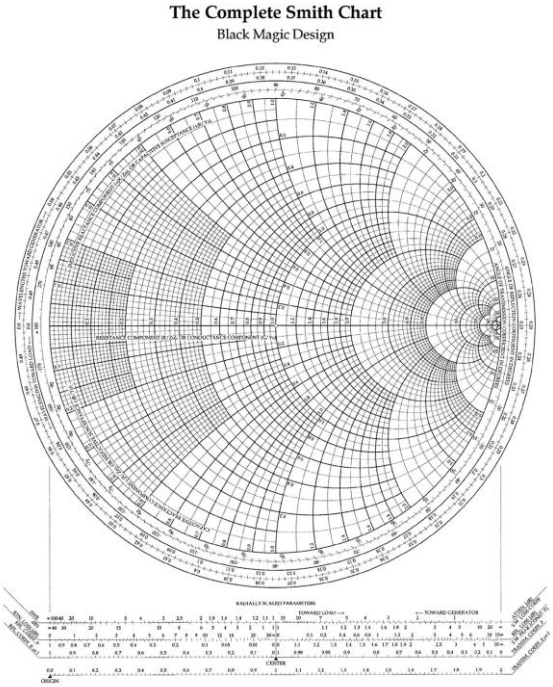
Lecture Objective

- By the end of this lecture, you will :
  - Understand how the Smith chart is obtained and its usefulness for transmission line calculations.

The Smith Chart

- The Smith chart was invented by Philip Smith at Bell Telephone's Radio Research Lab during the 1930s.
- An article authored by Smith on the development was initially rejected for publication.
- The Smith chart enables us to solve transmission line problems graphically.
- The Smith chart enhances a deep intuitive understanding of solutions to transmission line problems.



What is the Smith Chart?

- The Smith chart is a plot of the complex reflection coefficient plane (imaginary Γ versus real Γ) with a grid showing the mapping of certain normalized impedance (or admittance) values onto this plane.
- In other words, the Smith chart is a mapping between the complex normalized impedance (or admittance) plane and the complex reflection coefficient plane.

Mapping from Impedance to Reflection Coefficient

- The mapping from an impedance value to a reflection coefficient value is defined by

$$\boxed{\Gamma = \frac{Z - Z_0}{Z + Z_0} = \frac{z - 1}{z + 1}}$$

- Where the normalized impedance is

$$z = Z / Z_0$$

Mapping from Reflection Coefficient to Impedance

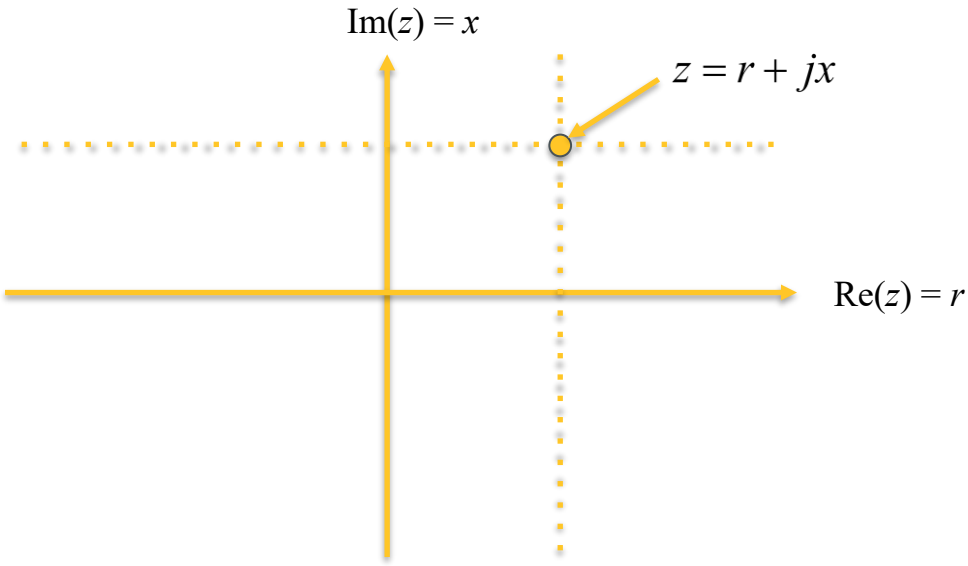
- The inverse mapping from a reflection coefficient value to a normalized impedance value is defined by

$$\boxed{z = \frac{1+\Gamma}{1-\Gamma}}$$

What is the meaning of 'Z_0'?

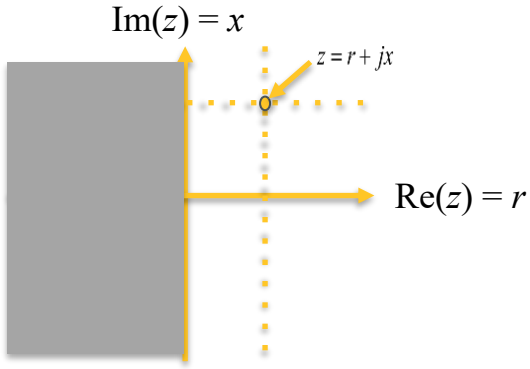
- Previously, we used the symbol 'Z_0' to denote the characteristic impedance of a particular transmission line.
- In the present context, it is more precise to say that the symbol 'Z_0' denotes a reference impedance (commonly, 50 ohms).
- In the given system, various transmission lines might have characteristic impedances equal to the reference impedance, but this need not be so.

Complex (Normalized) Impedance Plane



2. Complex (Normalized) Impedance Plane

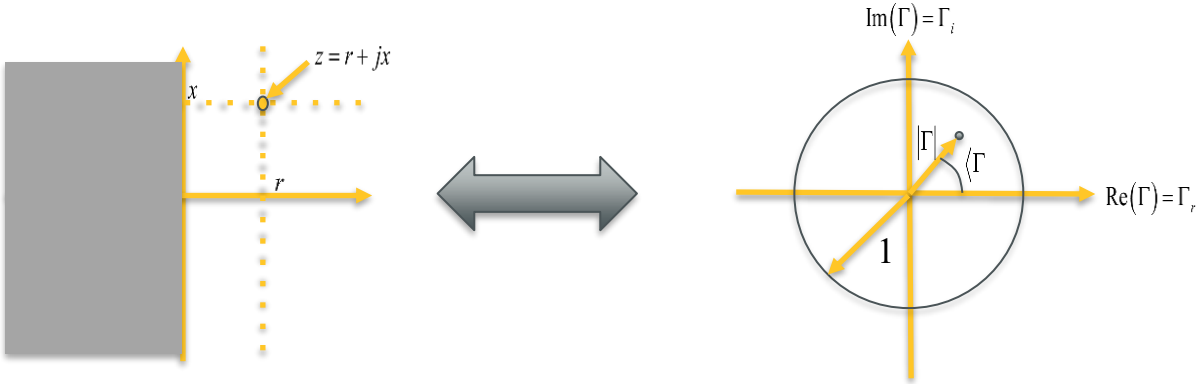
- For <u>passive</u> circuits: $r \geq 0, -\infty < x < \infty$



We need only consider points in the right half of the impedance plane. For these points:

$$|\Gamma| \leq 1$$

2. Mapping from Impedance to Reflection Coefficient



Mapping from Impedance to Reflection Coefficient Using the Parametric Equations

- In terms of the real and imaginary coordinates in each plane, the mapping can be written as

$$\left(\Gamma_r - \frac{r}{1+r}\right)^2 + \Gamma_i^{\,2} = \left(\frac{1}{1+r}\right)^2$$

$$\left(\Gamma_r - 1\right)^2 + \left(\Gamma_i - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2$$

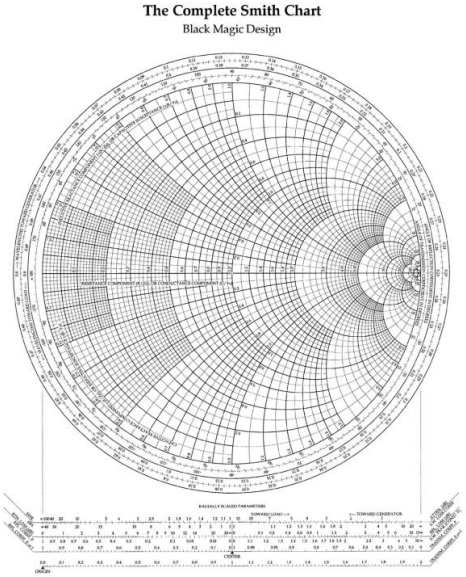
Notice both equations are in the form of

$(x\text{-}a)^2 + (y\text{-}b)^2 = R^2$

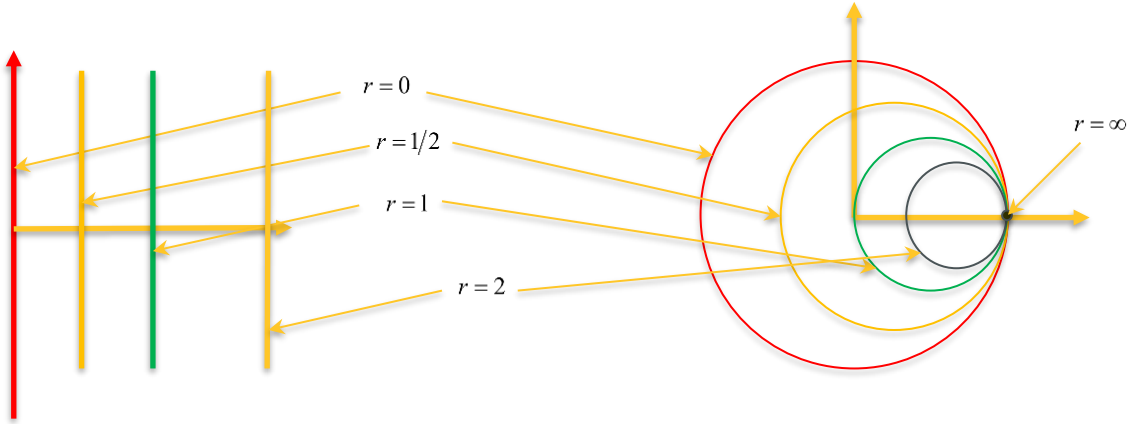
These equations represent two families of circles in the complex reflection coefficient plane.

3. Mapping from Impedance to Reflection Coefficient

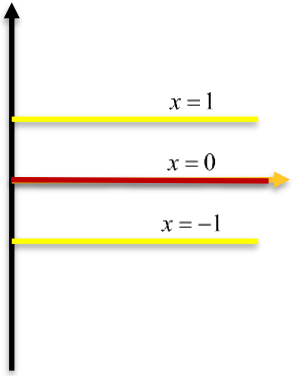
- These parametric equations are the ones that are used to make the circles representing the impedance coordinates on the Smith chart.

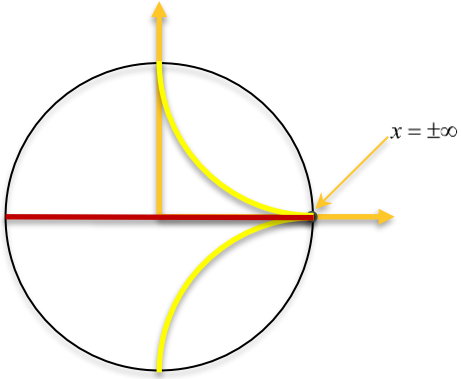


Generating the Smith Chart



2. Generating the Smith Chart



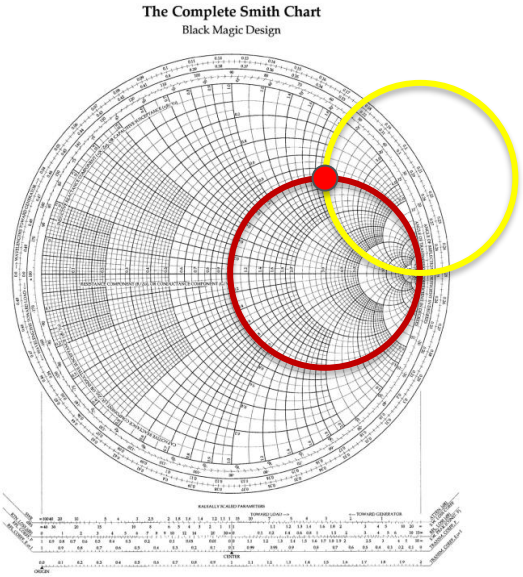


It is interesting to note that the circles (lines) of constant normalized resistance and the circles (lines) of constant normalized reactance are everywhere perpendicular to each other in both reflection coefficient and impedance planes.

4. Mapping from Impedance to Reflection Coefficient

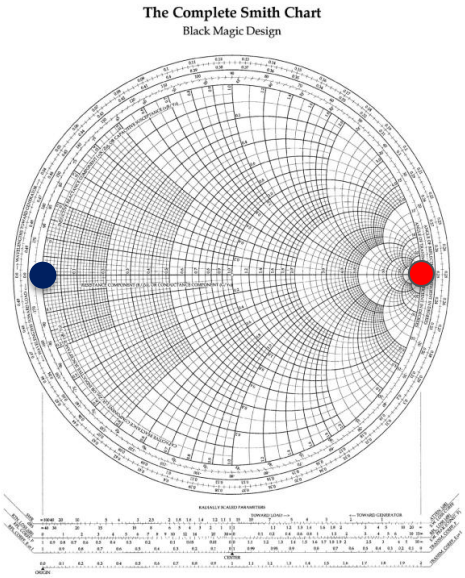
- To map a point in the complex (normalized) impedance plane onto the Smith chart, we can determine the intersection of the corresponding circles.
- For example

$$z = 1 + j2$$



5. Mapping from Impedance to Reflection Coefficient

- The point $\Gamma = 1$ represents an open circuit ($z \rightarrow \infty$).
- The point $\Gamma = -1$ represents a short circuit ($z = 0$).



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